

Total No. of Questions : 8]

SEAT No. :

PD-4280

[Total No. of Pages : 2

[6403]-77

T.E. (Electronics / Electronics & Telecommunication Engineering)

DATA BASE MANAGEMENT

(2019 Pattern) (Semester - V) (304183)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat Diagram must be drawn wherever necessary.
- 3) Assume Suitable data if necessary.
- 4) Figure to the right indicates full marks.

- Q1) a)** Explain the use of constraints in SQL. List and explain any three constraints with example. **[7]**
- b)** What are different types of joins in SQL? Explain any one with suitable example. **[6]**
- c)** Explain with example any two STRING functions used in SQL. **[5]**

OR

- Q2) a)** What are some common clauses used with SELECT query in SQL? Explain any two with example. **[7]**
- b)** What is Cursor? Explain Implicit and Explicit cursor in PL/SQL with suitable example. **[6]**
- c)** Explain the use of TRUNCATE, DELETE and DROP statements in SQL? Explain any one with example. **[5]**

- Q3) a)** Explain briefly different states of transaction with a neat state transition diagram. **[6]**
- b)** Define transaction. State and explain in brief the ACID properties of a transaction? **[5]**
- c)** When deadlock occurs? How to prevent them? How to recover if dead lock occurs. **[6]**

P.T.O.

OR

- Q4)** a) Explain the term i) Conflict Serializability ii) View Serializability [6]
b) What is Concurrent execution of transaction? Explain any two concurrency problems occurs during Concurrent execution of transaction with examples? [5]
c) Explain Commit and Rollback operations of Transactions. [6]

- Q5)** a) What are performance parameters for parallel database? [6]
b) Explain virtualization on Multicore processors with neat schematic diagram. [6]
c) What is Parallelism? Explain Inter query parallelism. [6]

OR

- Q6)** a) List the parallel database architectures and explain any one architecture. [6]
b) Explain Oracle Database Architecture. [6]
c) Explain client server architecture with suitable example. [5]

- Q7)** a) Explain Homogeneous and Heterogeneous Distributed Databases. [6]
b) Explain Fragmentation and Replication in Distributed data storage. [6]
c) What is distributed transaction? Explain failure modes of distributed transaction. [6]

- Q8)** a) What is necessity of distributed database management system? What are the Factors Encouraging DDBMS. [6]
b) Explain two phase commit protocol in distributed database. [6]
c) Draw and explain client-server architecture for DDBMS. [6]

